Derivatives Securities - Lecture 6 Mechanics of Options Markets

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Types of options

- An option gives the option buyer the ***right, but not the obligation,*** to buy or sell a security to the option writer/seller
  - at a pre-specified price (***the strike price, K***)
  - at (or up to) a given date in the future (***the expiration date, T***)

- An option has **positive value**

- Option types
  - **Call option** – the option to ***BUY***
  - **Put option –** the option to **SELL**
  - **American** option can be exercised at any time prior to maturity
  - **European** option can be exercised only at maturity

Option positions

There are two sides to every option contract

Call:

- Long call (buying the call)
- Short call (selling/writing the call)

Put:

- Long put (buying the put)
- Short put (selling/writing the put)

Call option
Payoffs and Profits

A call option should always be exercised at expiration if $\boldsymbol{S_T \geq K}$

Position	Initial *Cost*	Terminal *Payoff*		*Profit*	
		$\boldsymbol{If\ S_T < K}$ (no exercise)	$\boldsymbol{If\ S_T \geq K}$ (exercise)	$\boldsymbol{If\ S_T < K}$ (no exercise)	$\boldsymbol{If\ S_T \geq K}$ (exercise)
Long call (buyer)	c	0	$S_T - K$	$-c$	$S_T - K - c$
Short call (writer)	$-c$	0	$K - S_T$	c	$K - S_T + c$

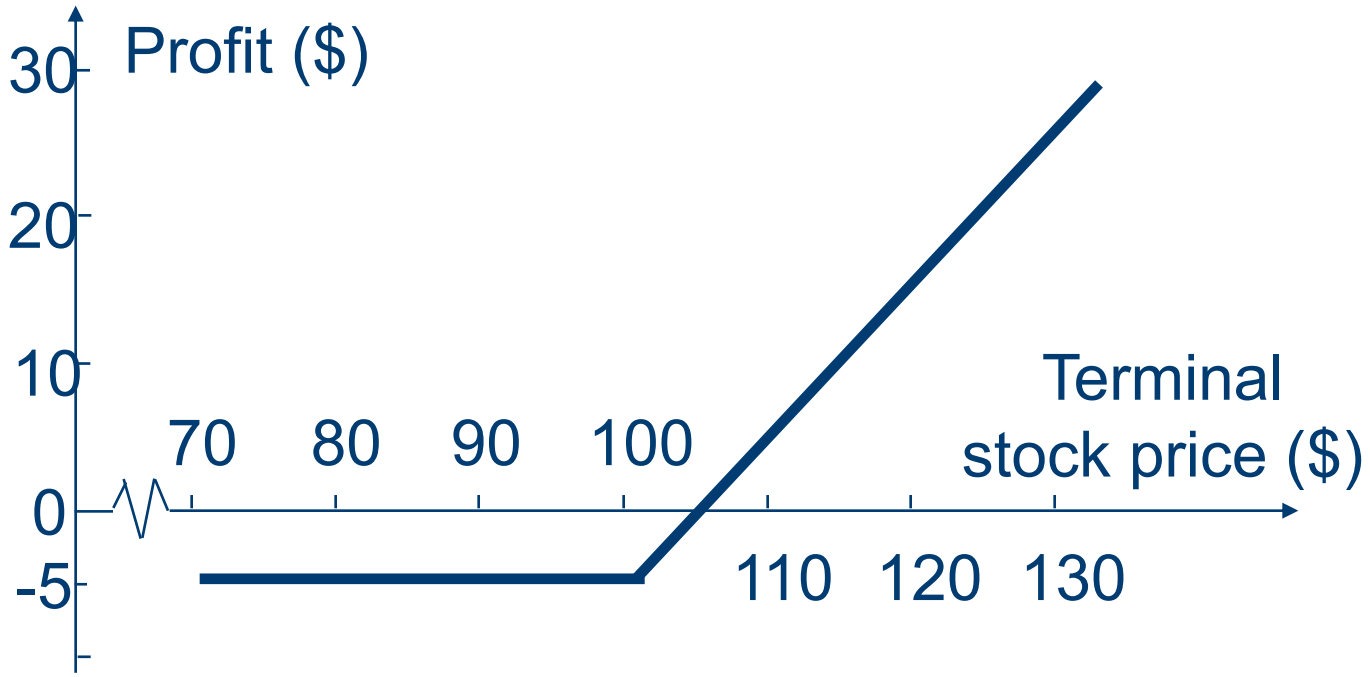
Break-even price – the terminal spot price that makes Profit=0

$$\boldsymbol{S_T^{Break-even} = K + c}$$

Long Call
Profit diagram

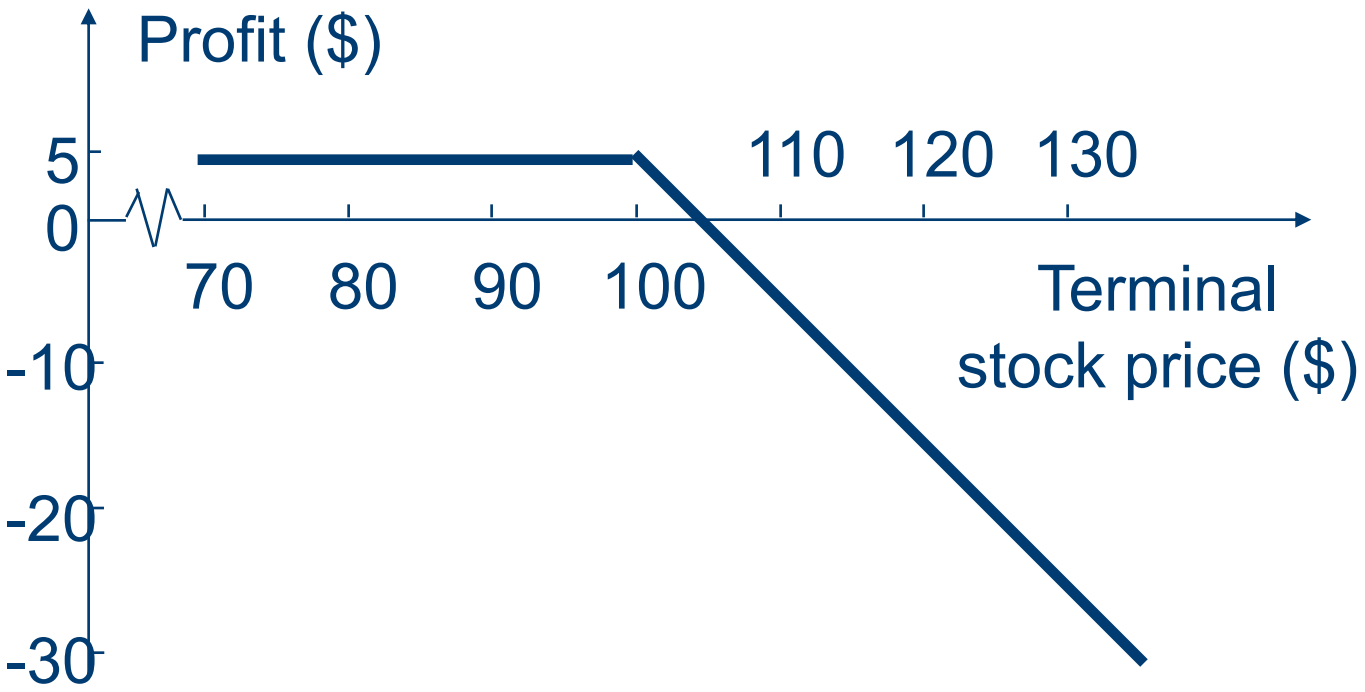
Profit from buying one European call option: option price = \$5, strike price = \$100.

$$S_T^{Break-even} = 105$$



Short Call (writing a call)
Profit diagram

Profit from writing one European call option: option price = $5, strike price = $100.



Put option
Payoffs and Profits

A put option should always be exercised at expiration if $S_T < K$

Position	Initial *Cost*	Terminal *Payoff*		*Profit*	
		$If\ S_T < K$ (exercise)	$If\ S_T \geq K$ (no exercise)	$If\ S_T < K$ (exercise)	$If\ S_T \geq K$ (no exercise)
Long put (buyer)	p	$K - S_T$	0	$K - S_T - p$	$-p$
Short put (writer)	$-p$	$S_T - K$	0	$S_T - K + p$	p

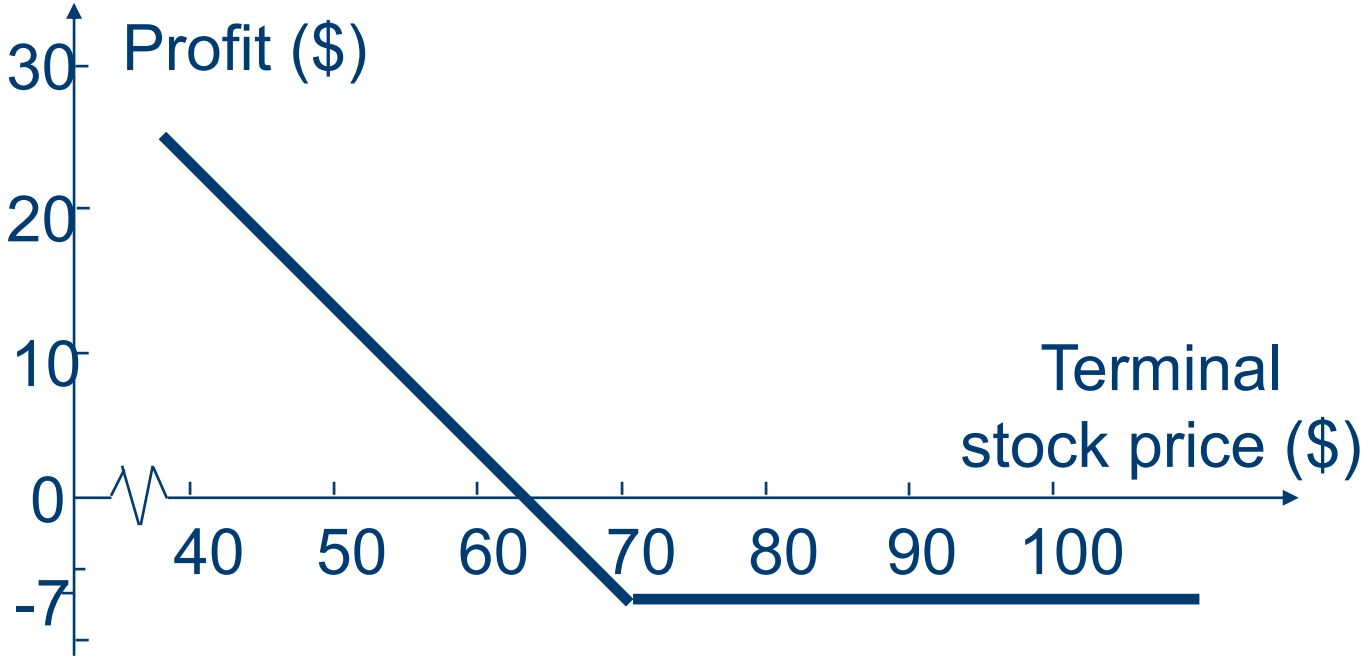
Break-even price – the terminal spot price that makes Profit=0

$$S_T^{Break-even} = K - p$$

Long Put
Profit diagram

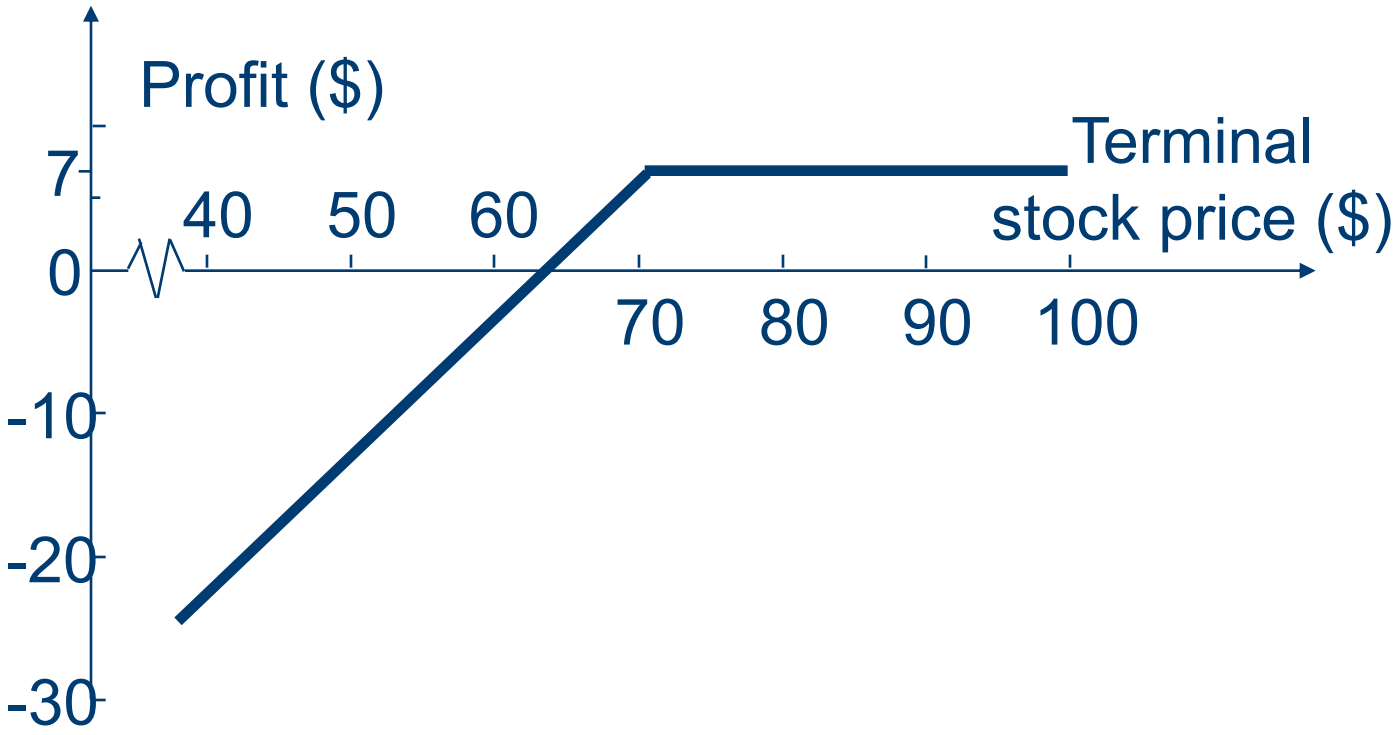
Profit from buying a European put option: option price = \$7, strike price = \$70

$$S_T^{Break-even} = \$63$$



Short Put (writing a put)
Profit diagram

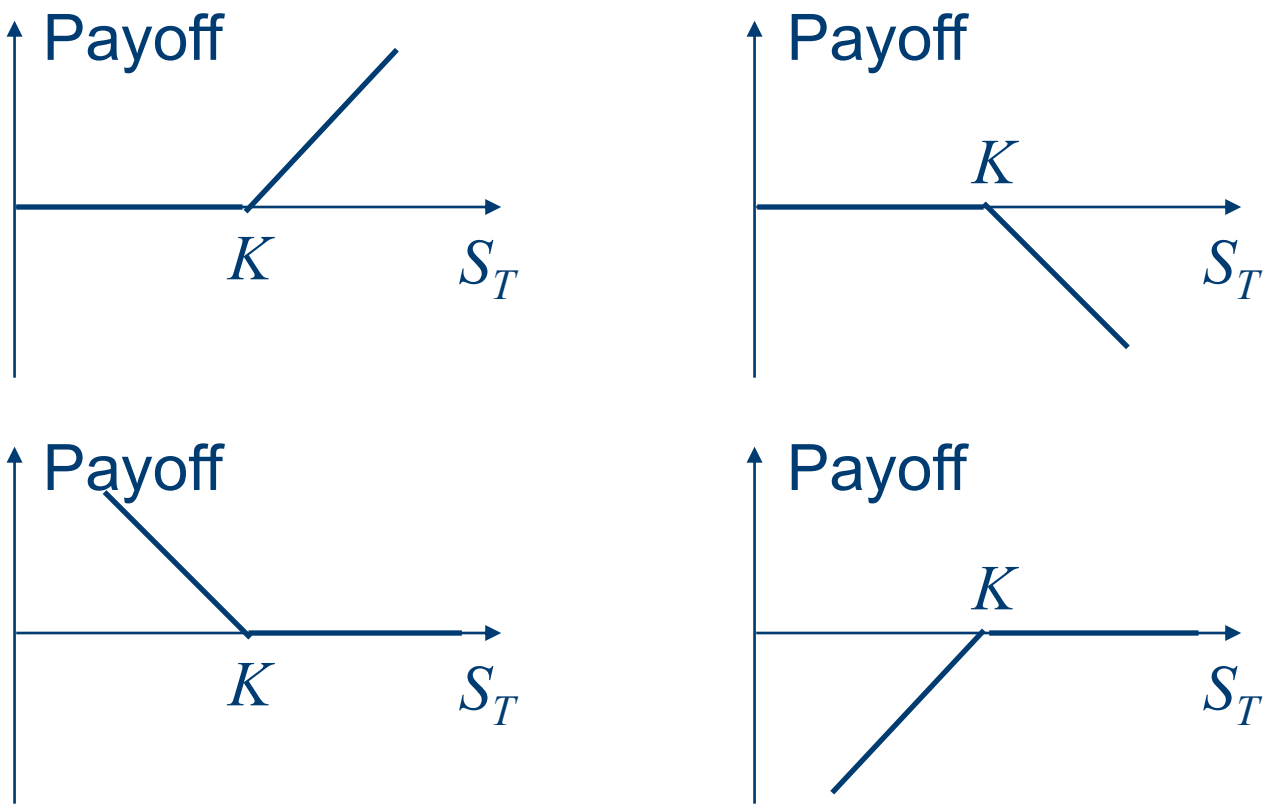
Profit from writing a European put option: option price = $7, strike price = $70



Payoffs from option positions

K = Strike price, S_T = Price of asset at maturity

What is the Option Position in Each Case?



Discuss

Explain carefully the difference between writing a put option and buying a call option.

Option value

The value of an option is determined by:

- the current spot price (S_t)
- the strike price K,
- the time to maturity
- interest rates
- dividends
- the option type (Call or put, American or European), and
- the dynamics of the underlying security (e.g., how volatile the security price is)

Option value

We can decompose the value of each option into two components:

Option Value = Intrinsic Value + Time Value

- **Intrinsic value** – the maximum of zero and the payoff if it were exercised immediately

  Call: $IV_t = \max(S_t - K, 0)$

  Put: $IV_t = \max(K - S_t, 0)$

- **Time value** - determined by time to maturity and the dynamics of the underlying security

Moneyness

Moneyness – determined by the strike price relative to the spot price

- **<u>At-the-money</u>** option – when the strike equals the spot (S_t=K)

- **<u>In-the-money</u>** - if the option has positive value if exercised right now (i.e. IV>0)
  Call: S_t>K
  Put: S_t<K

- **<u>Out-of-the-money</u>** - if the option has zero value if exercised right now (i.e. IV=0)
  Call: S_t<K
  Put: S_t>K

Note: Out-of-the-money options do not have intrinsic value, but they have time value.

Example 1

Consider a European call option on a stock index. The current index level (spot S_t) is 100. The option has a strike (K) of \$90 and a time to maturity of 1 year. The option has a current value (c_t) of \$14.

- *Is this option in-the-money or out-of-the-money?*
- *What's the intrinsic value for this option? What's its time value?*
- *What is the break-even spot price?*
- *If you buy this option, what's your payoff and profit if the index level reaches 110, 95, or 80 at expiration?*
- *If you write this option, what's your payoff and profit if the index level reaches 110, 95, or 80 at expiration?*

Underlying assets for exchange-traded options

Underlying	Typical Contract Size	Exercise Style	Settlement	Notes
Stocks	100 shares	American	Physical delivery of shares	Most common equity options
ETPs (e.g., ETFs)	100 shares	American	Physical delivery	Trade like stocks; track benchmarks
Foreign Currency	10,000 units *(1,000,000 for JPY)*	European	Cash	Traded on exchanges such as CME
Stock Indices	100 × index level	European *(S&P 100 is American)*	Cash	No delivery of component stocks
Futures Options	Based on futures contract size	American (typical)	Futures position upon exercise	Exercising creates a futures contract

Specification of Exchange-Traded Stock Options

American-style options to buy or sell 100 shares of a stock

Expiration dates:

- Expiration is the third Friday of the expiration month
- Trade on January, February or March cycle (every 3 months)
  - E.g. January cycle is Jan, Apr, Jul, Oct
- Before the third Friday, options trade for the current month, the following month and the next two months of the cycle
- After the third Friday, option trade for the following two months and the next two months of the cycle.
- LEAPS (long-term equity anticipation securities) – up to 39 months in the future; always third Friday of January
  - Used for long-term hedging or directional strategies

Example 2: *Options on General Motors stock are on a March, June, September, and December cycle. What options trade on (a) March 1, (b) June 30, and (c) August 5?*

Specification of Exchange-Traded Stock Options

Strike price spacing – usually spaced $2.5, $5 or $10 apart:

Stock Price (S)	Typical Strike Interval	Example
$5 – $25	$2.50	10, 12.5, 15, 17.5
$25 – $200	$5	50, 55, 60, 65
Above $200	$10	210, 220, 230

Option class: all options of the same type (calls or puts) on an underlying asset

- E.g. GM calls = one class; GM puts = another class

Option series: all the options of a given class with the same expiration date and strike price (i.e. a particular contract that is traded)

- E.g. GM June 60 Call = one option series; GM June 60 Put = another series

Specification of Exchange-Traded Stock Options

Position limits – maximum number of option contracts an investor can hold on one side of the market

- long call and short put are on the same side - bullish
- short call and long put are on the same side – bearish

Limits prevent excessive market influence or manipulation.

Exercise limit (usually equals the position limit) – maximum number of contracts that can be exercised in any period of 5 consecutive business days.

- Largest stocks - 250,000 contracts
- Smaller, less traded stocks - 200k, 75k, 50k or 25k contracts

Prevents market disruption from large exercises

Dividends and stock splits

- Exchange-traded options are normally **not adjusted** for **cash dividends**
  - Reflected in option prices but not terms

- The strike price and number of shares are **adjusted** for **stock dividends** and **stock splits**

- Suppose you own options with a strike price of K to buy (or sell) N shares:
  - If there is an n-for-m stock split
    - strike price is reduced to $K*(m/n)$
    - # of shares that can be bought (or sold) is increased to $N*(n/m)$

  - If there is a $x\%$ stock dividend
    - strike price is reduced to $K/(1+x/100)$
    - # of shares that can be bought (or sold) is increased to $N*(1+x/100)$

Dividends and stock splits
Example 3

A trader has a put option contract to sell 100 shares of a stock for a strike price of $60.
What is the effect on the terms of the contract of:

(a) A $2 cash dividend being declared

(b) A 5-for-2 stock split

(c) A 5% stock dividend being paid

Margin Requirements

Buying vs Writing Options

- Option buyers
  - Pay full premium
  - No margin required
- Option writers
  - Must post margin
  - Margin recalculated daily
  - Margin call if account falls below requirement

Why?

- Options already embed leverage.
- Naked writing creates large contingent liabilities.
- Daily margining protects the financial system.

A naked option position exists when an investor:
•writes (sells) an option, and
•does not hold the underlying asset or a hedging position.

Margin Requirements

- Margins are required when options are ***sold/written***.
- When a naked option is written the margin is the greater of:
  - A total of 100% of the proceeds of the sale (premium) plus 20 % of the underlying share price less the amount (if any) by which the option is out of the money
  - A total of 100% of the proceeds of the sale plus
    - 10% of the underlying share price for a call
    - 10% of the strike price for a put

Out-of-the-Money (OTM) Amount
Call: $\max(0, K - S)$
Put: $\max(0, S - K)$

- Note: the 20% is replaced by 15% for options on a stock index (it's less volatile)

Margin Requirements
Example 4

A trader writes five naked put option contracts, with each contract being on 100 shares. The option price is $10, the time to maturity is six months, and the strike price is $64.

(a) What is the margin requirement if the stock price is $58?

(b) How would the answer to (a) change if the rules for index options applied?

(c) How would the answer to (a) change if the stock price were $70?

(d) How would the answer to (a) change if the trader is buying instead of selling the options?

Trading

- More than 95% exchange-traded options are traded electronically

- Most exchanges use **market makers** to facilitate options trading (i.e. provide liquidity)
  - A market maker quotes both bid and ask prices when requested
  - The benefit of market making is the bid-ask spread
  - The exchange sets upper limits for the bid–offer spread based on the option price (e.g. no more than $0.75 for options priced between $10 and $20)
  - The risk of market making is market movements (counterparty/credit risk)
  - Market makers hedge their risk exposures by buying/selling stocks according to their option inventories.

Trading

Positions can be closed by taking the opposite position:

- Buy a call/put → close by selling the same call/put
- Write (sell) → close by buying

Why Traders Offset Positions

✓ lock in profits
✓ stop losses
✓ avoid exercise/assignment
✓ adjust portfolio risk

Most options positions are closed before expiration.

Situation	Effect on Open Interest
Buyer and seller opening new positions	increases by 1
One opens, one closes	no change
Both closing	decreases by 1

Other option-type products: Warrants

- Warrants are options issued by a corporation or financial institution, rather than traded on an options exchange.
  - Terms are set by the issuer
  - Exercised directly with the issuer
  - Often long-term instruments
- Warrants Outstanding
  - Quantity is fixed at the original issue
  - Changes only when warrants are exercised or expire
- Settlement
  - Holder exercises with the issuer
  - Issuer settles by:

    - delivering new shares (corporate warrants), or

    - settling per contract terms (financial institution warrants)

Other option-type products: Warrants

- Warrants Issued by Financial Institutions
  - May be written on commodities, currencies, or indices
  - Issuer typically creates a market for the warrants
  - Holder exercises through the issuing institution

- Corporate Warrants (on the firm's own stock)
  - Exercise results in new shares being issued
  - Increases shares outstanding → potential share dilution
  - Often issued with bonds to make them more attractive and lower borrowing costs

Other option-type products: Employee stock options

Employee stock options - call options issued by a company to its executives as a form of compensation

- Usually at-the-money when issued
- When exercised, the company issues more stock and sells it to the option holder for the strike price
- They become vested after a period of time (1 to 4 years)
- They cannot be sold

"If a company does not do better than its competitors but the stock market goes up, executives do very well from their stock options. This makes no sense" Discuss.

Other option-type products: Convertible bonds

Convertible bonds - bonds that can be exchanged for equity at certain times in the future according to a predetermined exchange ratio

- Equivalent to a bond with an embedded call option on the company's stock
- Very often callable, so that the issuer can force conversion at a time earlier than the holder might otherwise choose.

Other option-type products vs Exchange-traded options

- Warrants, employee stock options, and convertibles:
  - A predetermined number of options are issued
  - When calls are exercised, new stock is issued

- Exchange-traded options:
  - Number of options is not predetermined (as more investors take or close out positions, the number of outstanding options changes)
  - When calls are exercised, the company whose stock underlies the option is not involved (no new stock)